VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
B.TECH. MINOR IN INNOVATION AND ENTREPRENEURSHIP

COURSE STRUCTURE AND SYLLABUS
(Applicable for the batches admitted from the academic year 2022-2023)

V SEMESTER **R22**

Course Code	Title of the Course	L	T	P/D	CH	C
22MC1IE301	Introduction to Innovation and Entrepreneurship	3	0	0	3	3
22MC2IE301	Customer-centric Innovation and Business Modelling through Design Thinking	0	0	3	3	1.5
Total		**3**	**0**	**3**	**6**	**4.5**

VI SEMESTER **R22**

Course Code	Title of the Course	L	T	P/D	CH	C
22MC1IE302	Startup Finance	3	1	0	4	4
Total		**3**	**1**	**0**	**4**	**4**

VII SEMESTER **R22**

Course Code	Title of the Course	L	T	P/D	CH	C
22MC1IE401	Market Research and Sales	3	0	0	3	3
22MC2IE401	Business Analytics Laboratory	0	0	3	3	1.5
Total		**3**	**0**	**3**	**6**	**4.5**

VIII SEMESTER **R22**

Course Code	Title of the Course	L	T	P/D	CH	C
22MC1IE402	Startup Management and Strategy	3	0	0	3	3
22MC1IE403	Digital Marketing					
22MC1IE404	Social Entrepreneurship					
22MC1IE405	Ethics and Law for Entrepreneurs					
22MP1IE401	Mini – Project	0	0	4	4	2
Total		**3**	**0**	**4**	**7**	**5**

L – Lecture T – Tutorial P – Practical D – Drawing CH – Contact Hours/Week
C – Credits SE – Sessional Examination CA – Class Assessment ELA – Experiential Learning Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation LR – Lab Record
CP – Course Project PE – Practical Examination

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (I & E) V Semester

(22MC1IE301) INTRODUCTION TO INNOVATION AND ENTREPRENEURSHIP

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To identify innovations in existing solutions, understand innovation through design thinking, and appreciate innovation as a contributor for intellectual property
- To embark the path of entrepreneurship equipped with an understanding of entrepreneurial mindset and support schemes in India
- To ideate innovative solutions for problems using design thinking as well engineering knowledge
- To develop a fundamental market awareness and business modelling

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Understand innovation and use design thinking to create innovations
CO-2: Develop innovation process, manage innovation, and generate IP if any
CO-3: Understand and develop the mindset and skills for entrepreneurship
CO-4: Convert a technology innovation into a business solution understanding markets and customers
CO-5: Design a business model and understand fundamentals of business strategy

UNIT-I:
Introduction to Innovation: Innovation and Creativity, Motivation, Innovation in Current Environment, Analysing the Current Business Scenarios and Trends, Types of Innovation, Challenges of Innovation, Sources of Innovation, Idea Management System, Divergent vs. Convergent Thinking
Design Thinking for Innovation: Customer-centric Approach—Motivation and Merits; Desirability, Feasibility, Viability, Scalability; Design Thinking for Innovation Use cases: Relevant Challenges, Applications to Business Models, Sales, Marketing, and Strategy Management.

UNIT-II:
Innovation Management and Intellectual Property: Innovation Life Cycle, Roles in Innovation Process, Innovation Management, Experimentation in Innovation Management, Participation for Innovation, Co-creation for Innovation, Prototyping to Incubation, Innovation and Creation of IP, Types of IP Rights, Patenting in India.

UNIT-III:
Introduction to Entrepreneurship: Entrepreneurship vs Innovation, Types of Entrepreneurship, Human side of entrepreneurship, Myths & Realities about entrepreneurship, Entrepreneurial qualities and how to develop them, Initiatives by the Government of India to promote entrepreneurship, Introduction to the Lean Startup methodology

UNIT-IV:
Understanding Needs, Customers, and Markets

Identifying a problem and a need, Ideation: why and how, Developing a Value Proposition, Unique Value Proposition, Value Chain Analysis, How-Might-We Statements, Validation, Market research (Significance, Impact, who and how, Segmentation), Reasons for Failure of Startups

UNIT-V:

Business Model and Beyond: Business model, Business Model Canvas (BMC), Lean Canvas (LC), Product-Market Fit (PMF), Kano Model, Action Priority Matrix, Value-Ease Matrix, Go-to-Market (GTM) Strategies, Blue Ocean Strategy PESTLE Analysis, Porter's 5 forces model, Competitor landscape, SWOT analysis

TEXT BOOKS:

1. The Lean Startup: How Today's Entrepreneurs Use Continuous Innovation to Create Radically Successful Businesses, Eric Ries, Penguin Portfolio, 2011
2. Entrepreneurship Simplified: From Idea to IPO, Ashok Soota, S. R. Gopalan, India Portfolio, 2021
3. Entrepreneurship: New Venture Creation, David H. Holt, Pearson Education India, 2016

REFERENCES:

1. The Manual for Indian Start-Ups, Vijay Kumar Ivaturi et al., Penguin Enterprise, 2020
2. Engineered in India: From Dreams to Billion-Dollar Cyient, B. V. R. Mohan Reddy, Penguin Business, 2022
3. Before You Start Up: How to Prepare to Make Your Startup Dream a Reality, Pankaj Goyal, Fingerprint! Publishing, 2017
4. How to Start Business in India: Idea to Implementation, Dibyendu Choudhury, Notion Press, 2023
5. Entrepreneurship, Robert D. Hisrich et al., McGraw-Hill, 11^{th} Edition, 2020

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (I & E) V Semester

(22MC2IE301) CUSTOMER-CENTRIC INNOVATION AND BUSINESS MODELLING THROUGH DESIGN THINKING

TEACHING SCHEME		
L	T/P	C
0	3	1.5

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To use design thinking approach and methods from problem identification to business modelling
- To innovate customer-centric solutions for identified problems, identifying value proposition
- To create and present a business model using business model canvas or lean canvas
- To use different canvases, templates, and online tools in the process of business model creation

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Use design thinking approach to identify and validate problem, identify users and customers and understand their jobs to be done to deduce value proposition for them
CO-2: Use suitable canvases and templates to frame problem, empathise with users, ideate, present value proposition, and create a valuable, usable, and possible innovative solution using suitable ideation techniques
CO-3: Create a business model using business model canvas or lean canvas and any other templates for finer representation
CO-4: Analyse business model using suitable techniques such as PESTLE, Kano model, Five Forces Model, and/or SWOT analysis
CO-5: Create pitch deck to present the problem and proposed solution highlighting innovative aspects therein as well as a fundamental market strategy

LIST OF EXERCISES / ACTIVITIES:

1. **WEEKS 1–3: Define & Empathise**
   a) Validation of idea represented using mind maps, concept maps, or any suitable template
   b) Problem framing using suitable canvas to choose parts of problem to solve, users to empathise with, and identify value proposition in principle, from a general problem expressed in 1(a)
   c) Identify Users and classify them for subsequent empathisation using Users Bull's Eye canvas if applicable, and create User persona(s) using suitable template
   d) Represent user empathisation using any of the following: empathy map, A-E-I-O-U framework, user interviews, surveys, value proposition canvas, and comparing with authentic data from public domain if any
   e) Write user needs statements in the format "As a …, I need/want/expect to … so that … because … when …" and classify primary, secondary, and latent needs

f) Iterate steps 1(c) through1(e) for further validation and fine-tuning, before writing the Specific Problem Statement using 5 W's

2. **WEEKS 4–6: Ideation & Prototype**
   a) Generate ideas using tools such as Brainstorming, SIT Methods, SCAMPER model, Storyboarding, Mind Maps and identify innovation and its value to users/ customers
   b) Compare ideas from 2(a) to see which needs from 1(e) are addressed for which users from 1(c)
   c) Create physical or digital models, mock-ups, wireframes, or other data visualizations to create solution prototypes based on the chosen idea(s) from 2(a) above
   d) Test and validate the prototype for valubility (desirability + viability), usability (desirability + feasibility), and possibility (feasibility + viability) with the help of users empathised in 1(d)
   e) Reconsider any SIT Methods for further innovation of tested and validated solution
   f) Identify relevant SDG attainment the final solution can contribute and make a report

3. **WEEKS 7–10: Business Modelling**
   a) Create a Customer Journey Map (CJM) or a Service Experience Cycle based on the chosen problem (1) and solution (2) above.
   b) Fill the Business Model Canvas or Lean Canvas considering desirability, feasibility, and viability
   c) Use any additional canvases such as Service Innovation Canvas, Social Business Model Canvas, Ethics Canvas, Data Ethics Canvas, Opportunity Canvas, and/or Project Canvas to represent additional aspects of the problem and/or solution

4. **WEEKS 11–13: Business Model Analysis**
   a) Apply PESTLE analysis, Kano Model, Porter's Five Forces, SWOT analysis to the business model
   b) Explore GTM strategies including Blue Ocean Strategy
   c) Create a business idea pitch deck and present

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (I & E) VI Semester

(22MC1IE302) STARTUP FINANCE

TEACHING SCHEME		
L	T/P	C
3	1	4

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:
- To understand the role and importance of Entrepreneurial Finance
- To provide conceptual knowledge on new ventures, startups, and evaluation
- To impart knowledge of various aspects of financial planning for startups
- To provide an understanding of various aspects of venture valuation
- To discuss the aspects of financing growing ventures

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Explain startup financing vis à vis the venture lifecycle of a company
CO-2: Assess first-round financing sources and their performance evaluation
CO-3: Plan and review financial aspects throughout the venture's life cycle
CO-4: Evaluate the mechanics and apply methods of Venture Valuation
CO-5: Elucidate the importance of various sources of venture capital financing

UNIT-I:
Finance Fundamentals:
Introduction: Principles of Entrepreneurial Finance; Role of Entrepreneurial Finance; **Fundamental Definitions:** Bootstrapping, Angel Investment, Venture Capital, Equity, Valuation, Dilution, Public Offering, Term Sheet, Cap Table, Burn Rate, Runway
Venture Lifecycle: Successful Venture Lifecycle; Financing through Venture Lifecycle; Lifecycle Approach for Teaching Entrepreneurial Finance
Screening Venture Opportunities: Pricing/ Profitability Considerations, Financial/ Harvest Considerations, Financial Plans and Projections

UNIT-II:
Organizing and Operating the Venture
Financing a New Venture: Seed, Startup and First Round Financing Sources, Financial Bootstrapping, Business Angel Funding, First Round Financing Opportunities.
Preparing and Using Financial Statements: Obtaining and Recording the Resources to Start and Build a New Venture, Asset and Liabilities and Owners Equity in Business, Sale Expenses and Profits Internal Operating Schedules, Statement of Cash Flows, Operating Breakeven Analysis, Evaluating Operating and Financial Performance using Ratio Analysis.

UNIT-III:
Financial Planning: Financial Planning throughout the Venture's Lifecycle, Short Term Cash Planning Tools, Projected Monthly Financial Statements
Types and Costs of Financial Capital: Implicit and Explicit Financial Capital Costs, Financial Markets, Determining the Cost of Debt Capital, Investment Risk, Estimating the Cost of Equity Capital, Weighted Average Cost of Capital

UNIT-IV:

Venture Valuation: Valuing Early-stage Ventures, Venture Worth, Basic Mechanics of Valuation, Developing the Projected Financial Statements for a Discounted Cash Flow Valuation, Accounting Vs Equity Valuation Cash Flow

Venture Capital Valuation Methods: Basic Venture Capital Valuation Method, Earnings Multiplier and Discounted Dividends

UNIT-V:

Financing for the Growing Venture: Professional Venture Capital, Venture Investing Cycle, Determining the Fund Objectives and Policies, Organising the New Fund, Soliciting Investments in the new Fund, Capital Call, Conducting Due-diligence and Actively Investing, Arranging Harvest or Liquidation

Other Financing Alternatives: Facilitators, Consultants and Intermediaries, Banking and Financial Institutions, Foreign Investors, State and Central Government Financing Programmes, Receivables Lending and Factoring, Mortgage Lending, Venture Leasing

TEXT BOOKS:

1. Entrepreneurial Finance, Leach J. C., Melicher R. W., Cengage Learning, 2023
2. First 1000 Days of Startup: Decoding Finance for Entrepreneurs, Pathak S. K., Notion Press, 2024
3. How to Raise Startup Funding in India, Ghuman K., Makkar S., Bluerose, 2024

REFERENCES:

1. Startup Finance 360°: A Founder's Guide to Startup Finance, Saria R., Zebra Learn, 2023
2. Financial Modeling Handbook - The Step-by-Step Guide to Building your First Financial Model & Value Companies from Scratch, Zebra Learn, 2023
3. Financing the Entrepreneurial Venture: A Casebook, Gompers P. A., Anthem Press, 2024
4. Entrepreneurial Finance: Strategy, Valuation, and Deal Structure, Smith J. K., Smith R. L., Notion Press, 2023
5. The Art of Startup Fundraising, Cremades A., John Wiley & Sons, 2016

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (I & E) VII Semester

(22MC1IE401) MARKET RESEARCH AND SALES

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To equip with fundamental and advanced concepts of market research and sales strategies in the digital era
- To develop the ability to apply various market research methods, including traditional and AI-driven techniques, to real-world business scenarios
- To enhance understanding of consumer behavior, segmentation, and targeting in the Indian and global markets
- To provide knowledge of effective sales strategies, customer relationship management, and AI-powered sales automation
- To leverage data-driven insights for improving sales performance and decision-making

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Conduct comprehensive market research using both qualitative and quantitative techniques, including AI-driven approaches
CO-2: Analyze consumer behavior and segment markets for targeted sales strategies
CO-3: Design and implement effective sales strategies, leveraging CRM and digital platforms
CO-4: Utilize sales forecasting and data analytics to improve business performance
CO-5: Understand ethical considerations and legal frameworks governing market research and sales in India

UNIT-I:
Fundamentals of Market Research
Introduction to Market Research: Definition, Importance, Applications
Types of Market Research: Primary vs. Secondary, Qualitative vs. Quantitative; Research Design and Sampling Methods
Data Collection Techniques: Surveys, Interviews, Focus Groups
Role of AI and Big Data in Market Research, Case Studies of Market Research in Indian Startups

UNIT-II:
Consumer Behavior and Market Segmentation: Understanding Consumer Psychology and Buying Behavior; Market Segmentation, Targeting, and Positioning (STP Model); Cultural, Social, Personal, and Psychological Influences on Buying Decisions; Customer Journey Mapping and Experience Design; Generative AI in Customer Insights and Personalized Marketing; Indian Consumer Trends and Preferences: A Data-Driven Perspective.

UNIT-III:
Sales Strategies and Techniques

Fundamentals of Sales: B2B vs. B2C Sales; Sales Funnel and Customer Relationship Management (CRM)
Digital Sales Channels: E-commerce, Social Selling, AI Chatbots Persuasive Selling Techniques and Negotiation Skills; Sales Automation and AI-powered Lead Generation; Case Studies of Indian and Global Sales Success Stories

UNIT-IV:
Data-Driven Decision Making in Sales
Sales Forecasting Techniques: Traditional vs. AI-based Predictive Analytics in Market Research and Sales; KPI Metrics for Sales Performance Measurement; Role of Business Analytics in Enhancing Sales Strategies; Pricing Strategies and Competitive Analysis; Case Studies on Data-driven Sales Growth

UNIT-V:
Ethical and Legal Considerations in Market Research and Sales
Ethical Issues in Market Research: Privacy, Bias, and Data Protection
Basic Overview of Legal Aspects of Sales in India: Consumer Protection Act, E-commerce Laws, Advertising Regulations, Fair-trade Practices,
Responsible AI and Ethical AI-driven Sales Techniques; Future Trends in Market Research and Sales
Case Studies: Ethical Dilemmas in Sales and Market Research

TEXT BOOKS:
1. Marketing Research: Text and Cases, Rajendra Nargundkar, McGraw-Hill Education, 2019
2. Unleashing the Power of Gen AI: A Guide to Innovative Marketing Strategies, Varun Mitra, Notion Press, 2024
3. Sales and Distribution Management: Decisions, Strategies and Cases, Richard R. Still, Cundiff W. Edward, Norman A. P. Govoni, Sandeep Puri, Pearson Education, 2024

REFERENCES:
1. Marketing Management: Indian Cases, Prachi Gupta et al., Pearson, 2024
2. Marketing Management, Philip Kotler et al., 16th Edition, Pearson Education, 2022
3. AI Powered Marketing, Shikhar Singh, McGraw-Hill Education, 2024
4. Customer Relationship Management: Concepts, Applications, and Technologies, Daniel D. Prior, Francis Buttle, Stan Maklan, Routledge, 2024
5. The Psychology of Selling, Brian Tracy, Harper Collins Leadership, 2022

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (I & E) VII Semester

(22MC2IE401) BUSINESS ANALYTICS LABORATORY

TEACHING SCHEME		
L	T/P	C
0	3	1.5

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:
- To equip with hands-on experience in essential business analytics techniques using industry-relevant tools
- To develop skills in data handling, visualization, and exploratory analysis for real-world business applications
- To introduce predictive modeling and machine learning for data-driven decision-making in marketing and sales
- To create automated reports and interactive dashboards for business insights
- To guide through a capstone project, applying analytics to solve a real-world business problem

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Utilize open-source business analytics tools to perform data processing, visualization, and analysis
CO-2: Apply predictive modeling and machine learning techniques to extract actionable business insights
CO-3: Design and develop interactive dashboards and automated reports for decision-making
CO-4: Implement AI-driven techniques such as clustering, classification, and forecasting for business applications
CO-5: Deliver a capstone project, demonstrating proficiency in real-world analytics problem-solving

LIST OF PROGRAMS / EXERCISES:

Introduction to Business Analytics & Tools
Outcome: Understanding fundamental concepts and working with analytics tools
Fundamentals: Descriptive, Predictive, and Prescriptive Analytics
Overview of any two Freeware Tools: Python (Pandas, Matplotlib, Seaborn), R, KNIME, RapidMiner, Tableau Public
Basic Data Handling: Importing, Cleaning, and Transforming Data

Data Visualization & Exploratory Data Analysis (EDA)
Outcome: Creating effective visualizations and identifying patterns in data
- Fundamentals of Data Visualization & Storytelling
- Hands-on with Python (Matplotlib, Seaborn) / R (ggplot2)
- Dashboard Creation using Tableau Public & Google Data Studio

Statistical Analysis and Predictive Modelling
Outcome: Applying statistics and predictive modeling techniques for business insights
- Descriptive Statistics, Probability Distributions
- Hypothesis Testing (t-test, Chi-square, ANOVA)

- Regression Models (Linear & Logistic) for Sales Forecasting & Customer Churn Prediction

Advanced Business Analytics – Clustering & Classification
Outcome: Implementing machine learning techniques for segmentation and classification
- K-Means Clustering, Hierarchical Clustering
- Decision Trees, Random Forest, Naïve Bayes for Customer Segmentation & Fraud Detection

Business Intelligence & Report Automation
Outcome: Developing dashboards and automating reports for data-driven decision-making
- Data Wrangling & Cleaning (Python: Pandas, NumPy)
- Creating Interactive Dashboards with Tableau / Google Data Studio
- Automating Reports for Business Insights

Capstone Project – Real-world Business Analytics Application
Outcome: Applying learned concepts to solve practical business challenges
- Identifying a Business Problem (Marketing, Finance, Operations, HR)
- Applying Data Collection, Preprocessing, Visualization, Modelling
- Developing Business Insights & Final Presentation

TEXT BOOKS:
1. Data Analytics, Anil Maheshwari, McGraw-Hill Education, 2021
2. Data Analytics with Python, Subhashini Sharma Tripathi, Wiley, 2020
3. A Tour of Data Science: Learn R and Python in Parallel, Vikram Tiwari, Nailong Zhong, Taylor & Francis, 2024
4. Business Analytics: Techniques and Applications, Gopal Sakarkar, Wiley, 2022

REFERENCES:
1. Data Science from Scratch: First Principles with Python, Joel Grus, O'Reilly, 2019
2. The Elements of Statistical Learning, Trevor Hastie, Robert Tibshirani, Jerome Friedman, Springer, 2017
3. R for Data Science, Hadley Wickham, Garrett Grolemund, O'Reilly, 2017
4. Data Strategy: How to Profit from a World of Big Data, Analytics and Artificial Intelligence, Bernard Marr, Kogan Page, 2021

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (I & E) VIII Semester

(22MC1IE402) STARTUP MANAGEMENT AND STRATEGY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To differentiate startup strategy from corporate strategy, with emphasis on decision-making under uncertainty and resource constraints
- To introduce clearly defined startup business model typologies, revenue models, and unit-economic drivers relevant to early-stage ventures
- To familiarise students with founder-level organisational, governance, and strategic control choices, including co-founder alignment and board oversight
- To enable formulation of growth, pricing, and go-to-market strategies using established startup metrics and execution frameworks
- To create awareness of Indian regulatory, compliance, ethical, and sustainability considerations, positioning compliance as a source of strategic advantage

COURSE OUTCOMES: After completion of course, the student should be able to

CO-1: Differentiate startup strategy from corporate strategy and apply structured strategy frameworks

CO-2: Identify, classify, and analyse defined business model typologies and unit-economic drivers

CO-3: Apply founder-level organisational, governance, and control mechanisms in startups

CO-4: Formulate growth, pricing, and go-to-market strategies using clearly defined metrics

CO-5: Demonstrate awareness of regulatory, compliance, ethical, and sustainability requirements in India

UNIT-I:

Overview of Strategic Inflection and Tools

Startup Strategy Overview: Startup vs Corporate Strategy; Strategy Formulation; Strategy Choice Cascade; Wardley Mapping

Strategy Approaches Overview: 3-Box Thinking; Blue Ocean Strategy; Lean Startup Management; Value Innovation through Strategy Canvas; Blitzscaling

UNIT-II:

Founder, Organisation & Governance

Founder's Mindset: Founder vs Professional Manager; Opportunity chasing vs Enterprise building; 3-box Thinking; Ethical Leadership concepts; Founder-Problem Fit and Founder-Market Fit Overview

Culture-driven Organisational Design: Flat, Agile/ Cross-functional, Holacracy, ambidextrous organisations

Overview of Legality-driven Organisational Structures: Sole Proprietorship, Partnership, LLP, Private Limited Company

Co-founder Agreements Basics: Roles, Vesting, IP Assignment, Decision rights, Dispute resolution, Equity split principles; Template Sources; Board structure; Founder-led governance; Awareness of Basic term-sheet elements.

UNIT-III:

Business Models, Revenue Models, and Strategic Analytics

Business Model Typologies: VTDF; 3-box thinking; transitional models; revenue streams matrix; Product-based, Service-based, Platform-based, Marketplace, Subscription, Usage-based, Hybrid models

Revenue Stream Typologies: One-time sales, Recurring revenue, Transaction fee, Licensing, Freemium

Unit Economics: Customer Acquisition Cost (CAC), Lifetime Value (LTV), Contribution Margin, Payback Period, Break-even Analysis

Pricing Strategies Overview: Cost-plus pricing, Value-based pricing, Tiered pricing, Usage-based pricing

Pivot Frameworks: Speed–Reversibility Matrix; Defensibility and Moats – Brand, Network Effects, Switching Costs, Regulatory Moats

Learning Cycles Overview: PDCA vs DMAIC – contextual application in startups; BADIR – hypothesis-driven analytics from business question to recommendations

UNIT-IV:

Growth, Scaling & Go-to-Market Strategy

Growth Strategies: Business Development vs. Sales; Business Scaling Overview; Organic vs Inorganic growth; Minimum Viable Audience (MVA)

Go-to-Market (GTM) Strategy Overview: Target segment, Channel choice, Pricing, Positioning

Digital Marketing Funnel Overview: Awareness, Acquisition, Activation, Retention, Revenue, Referral (AARRR, or "Pirate Metrics")

Execution Frameworks Overview: OKR, OGSM Framework, Kanban Diagram, Bud Caddell Diagram, MoSCoW, Risk Registers, ERRC Matrix

UNIT-V:

Compliance, Sustainability & Long-term Advantage

Regulatory Awareness Overview: MCA Filings; GST Registration and Returns; DPIIT Startup India recognition and benefits; DPDP Act Awareness; Compliance Calendars Investment Readiness: Due Diligence; Investment Readiness Levels

Impact & Sustainability: Triple Bottom Line (People, Planet, Profit); CSR obligations for growing firms (awareness); Compliance as a source of competitive advantage

TEXT BOOKS:

1. Entrepreneurship Simplified: From Idea to IPO, Soota, Ashok & Gopalan, S. R. India Portfolio (Penguin Random House), 2021
2. Strategic Management, Kazmi, Azhar; Kazmi, Adela, McGraw-Hill Education, 2020
3. The Manual for Indian Start-Ups, Ivaturi, Vijay Kumar et al., Penguin Random House India, 2017

4. Business Strategy Essentials You Always Wanted To Know: A Beginner's Guide to Strategic Management, Porter's Five Forces, Industry Life Cycles & Execution, Daum, Callie, Vibrant Publishers, 2020

REFERENCES:

1. Startup Management, Rothaermel, Frank T., McGraw-Hill, 2023
2. The Start-Up Code: Taking Your Company from Seed to Scale, Bansal, Mukesh, Penguin Viking, 2024
3. Strategic Management, Cherunilam, Francis, Himalaya Publishing House, 2021
4. Startups of Bharat: Stories of India's Million-Dollar Founders Under Thirty, Arora, Aditya; Pasricha, Surya, Penguin Business, 2021
5. The Startup Spirit: A New-Age Blueprint for Entrepreneurs, Seshadri, Sridhar; Iyer, Shreeram, Rupa Publications, 2025

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (I & E) VIII Semester

(22MC1IE403) DIGITAL MARKETING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand digital marketing as a strategic growth lever for startups and technology-driven ventures
- To familiarise with digital platforms, channels, and customer journey concepts relevant to startups
- To analyse customer engagement and digital marketing performance using metrics and analytics
- To examine ethical and legal dimensions of digital marketing in the Indian context

COURSE OUTCOMES: After completion of course, the student should be able to
CO-1: Explain digital marketing fundamentals and their relevance to startup business models
CO-2: Apply digital marketing channels and platforms for customer acquisition and reach
CO-3: Analyse digital customer journeys and engagement mechanisms
CO-4: Evaluate digital marketing performance using analytics and key metrics
CO-5: Identify ethical, legal, and responsible practices in digital marketing

UNIT-I:
Digital Marketing Foundations: Digital marketing ecosystem, evolution from traditional to digital marketing, relevance for startups and early-stage ventures, digital value propositions, integration with startup business models, overview of digital adoption and consumer behaviour in the Indian context

UNIT-II:
Digital Channels and Platforms: Search engine optimization and marketing, social media marketing, content marketing, email and mobile marketing, influencer marketing, platform selection criteria for startups, channel integration and coordination.

UNIT-III:
Customer Journey and Engagement: Digital customer journey mapping, acquisition–activation–retention frameworks, personalization approaches, engagement strategies, conversion concepts, brand building in digital environments

UNIT-IV:
Analytics and Performance Measurement: Key performance indicators, attribution models, web and social media analytics, campaign evaluation, budgeting considerations, return on investment analysis for startup marketing decisions

UNIT-V:

Ethics and Legal Aspects of Digital Marketing: Consumer protection, data privacy principles, ethical digital advertising practices, transparency, responsible use of customer data, regulatory considerations relevant to India.

TEXT BOOKS:

1. Digital Marketing, Seema Gupta, McGraw-Hill Education (India), 2018
2. Marketing 4.0: Moving from Traditional to Digital, Philip Kotler, Hermawan Kartajaya, Iwan Setiawan, Wiley India, 2017
3. Digital Marketing Analytics: Making Sense of Consumer Data in a Digital World, Chuck Hemann, Pearson Education India, 2014

REFERENCES:

1. Understanding Digital Marketing: Marketing Strategies for Engaging the Digital Generation, Damian Ryan, Calvin Jones, Kogan Page, 2016
2. Digital Marketing Excellence: Planning, Optimizing and Integrating Online Marketing, Dave Chaffey, Pearson Education, 2015
3. Digital Marketing in India: A Practical Approach, Paul Selva Raj, Notion Press, 2019
4. Digital Branding: A Complete Step-by-Step Guide to Strategy, Tactics and Measurement, Arvind Gupta, Sage Publications India, 2018
5. Online Consumer Behaviour, Jaishankar Ganesh, McGraw-Hill Education (India), 2015

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (I & E) VIII Semester

(22MC1IE404) SOCIAL ENTREPRENEURSHIP

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand social entrepreneurship as an approach to addressing societal challenges through enterprise
- To examine social innovation and hybrid organizational models
- To analyse sustainability and scaling strategies of social enterprises
- To appreciate the Indian social entrepreneurship ecosystem and institutional context

COURSE OUTCOMES: After completion of course, the student should be able to

CO-1: Explain the concept and evolution of social entrepreneurship

CO-2: Analyse social innovation processes and social enterprise business models

CO-3: Evaluate impact measurement and sustainability approaches in social ventures

CO-4: Assess scaling strategies and growth challenges of social enterprises

CO-5: Apply social entrepreneurship concepts to Indian development contexts

UNIT-I:

Introduction to Social Entrepreneurship: Concept and evolution of social entrepreneurship, distinction from commercial entrepreneurship, social value creation, stakeholder orientation, role in economic and social development

UNIT-II:

Social Innovation and Enterprise Models: Social innovation processes, inclusive and hybrid business models, mission–market alignment, balancing impact and financial viability.

UNIT-III:

Impact Measurement and Sustainability: Impact assessment concepts, qualitative and quantitative measures, financial sustainability, funding sources, resource mobilization challenges

UNIT-IV:

Scaling Social Enterprises: Growth and scaling strategies, partnerships, ecosystem support, institutional constraints, managing trade-offs between scale and impact

UNIT-V:

Indian Social Entrepreneurship Ecosystem: Sectoral focus areas such as health, education, agriculture and livelihoods, ecosystem actors, support institutions, contextual challenges in India.

TEXT BOOKS:

1. How to Change the World: Social Entrepreneurs and the Power of New Ideas, David Bornstein, Oxford University Press, 2007
2. Social Entrepreneurship, V. Kasturi Rangan, Lisa A. Chase, Sohel Karim, Harvard Business School Publishing, 2012
3. Entrepreneurship Development in India, K. Ramachandran, McGraw-Hill Education (India), 2014

REFERENCES:

1. Entrepreneurship Development, C.S.V. Murthy, Himalaya Publishing House, 2017
2. Building Social Business: The New Kind of Capitalism That Serves Humanity's Most Pressing Needs, Muhammad Yunus, PublicAffairs, 2010
3. The Fortune at the Bottom of the Pyramid: Eradicating Poverty Through Profits, C. K. Prahalad, Pearson Education India, 2009
4. Social Entrepreneurship, Johanna Mair, Jeffrey Robinson, Kai Hockerts (Eds.), Palgrave Macmillan, 2006
5. Social Innovation: Blurring Boundaries to Reconfigure Markets, Alex Nicholls, Oxford University Press, 2008

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. Minor (I & E) VIII Semester

(22MC1IE405) ETHICS AND LAW FOR ENTREPRENEURS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand legal frameworks and regulatory requirements relevant to startups in India
- To examine ethical challenges and decision-making in entrepreneurial contexts
- To develop awareness of compliance, governance, and risk management responsibilities
- To relate ethics and law to sustainable and responsible entrepreneurship

COURSE OUTCOMES: After completion of course, the student should be able to

CO-1: Identify legal foundations and business laws relevant to startups
CO-2: Explain contractual and regulatory obligations of entrepreneurial ventures
CO-3: Analyse ethical dilemmas faced by entrepreneurs
CO-4: Assess governance and compliance mechanisms in startups
CO-5: Relate ethics and law to long-term sustainability of enterprises

UNIT-I:
Legal Environment for Startups: Sources of Business Law; Forms of Business Organisation; Startup incorporation and compliance procedures; Contracts Overview; Introduction to intellectual property rights

UNIT-II:
Regulatory Compliance Framework: Labour laws, taxation basics, sector-specific regulations, compliance management, legal risk considerations for startups.

UNIT-III:
Ethics in Entrepreneurship: Ethical theories: Deontological Ethics, Utilitarianism; Ethical Ethical decision-making; Common ethical dilemmas in startups; Stakeholder responsibility and professional conduct; Need for Informed Consent; Privacy and Confidentiality in Non-disclosure Agreements (NDA) and Intellectual Property (IR) Rights (IPR); Accountability and Public-spiritedness vs. Profitability

UNIT-IV:
Governance and Accountability: Corporate Governance Principles, Leadership Responsibility, Transparency, Accountability mechanisms, Internal controls

UNIT-V:
Law, Ethics, and Sustainable Entrepreneurship: Responsible Business Practices, 3 C's of Business Ethics: Compliance, Contribution, and Consequences; Legal compliance for

long-term growth; Ethics as a driver of trust and sustainability; Using Data Ethics Canvas.

TEXT BOOKS:

1. Elements of Mercantile Law, N. D. Kapoor, Sultan Chand & Sons, 2019
2. Business Law, Bala Krishnamoorthy, McGraw-Hill Education (India), 2018
3. Corporate Governance and Business Ethics, M. Velayutham, McGraw-Hill Education (India), 2016

REFERENCES:

1. Business Ethics and Corporate Governance, P. Subba Rao, Himalaya Publishing House, 2018
2. Good Intentions Aside: A Manager's Guide to Resolving Ethical Problems, Laura L. Nash, Harvard Business School Press, 1993
3. Business Ethics: Concepts and Cases, Manuel G. Velasquez, Pearson Education, 2014
4. Legal Environment of Business, Robert A. Peterson, McGraw-Hill Education, 2015
5. Business Ethics: Managing Corporate Citizenship and Sustainability in the Age of Globalization, Andrew Crane, Dirk Matten, Oxford University Press, 2016